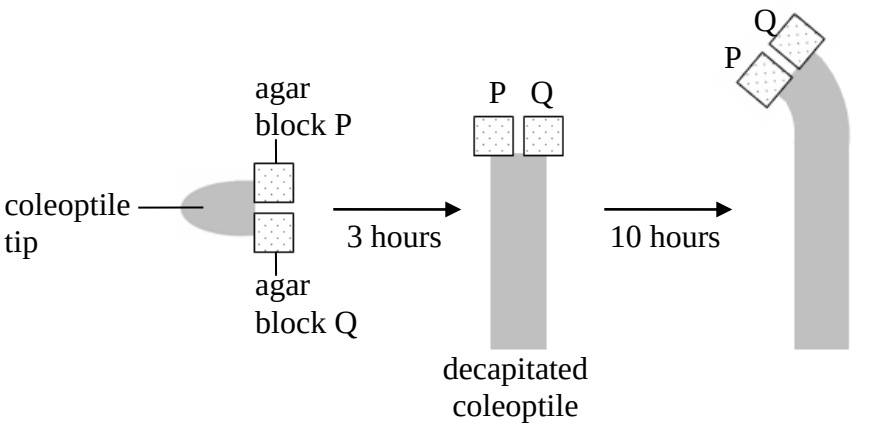
Structured questions

|!|BCELB0160000001|!| ★ **SI**

An experiment was performed on a number of coleoptiles in darkness. The coleoptiles were first decapitated. The tips were then placed horizontally. Two agar blocks (P and Q) were attached to each tip as shown in Figure I for three hours. Afterwards, the agar blocks were placed on the decapitated coleoptiles as shown in Figure II. The appearance of the coleoptiles after 10 hours was shown in Figure III.

Figure I **Figure II** **Figure III**



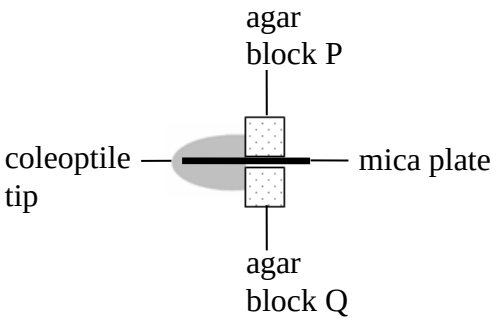
(a) Account for the appearance of the coleoptile shown in Figure III. (3 marks)

(b) If the two agar blocks in Figure II were put on a decapitated root, what would be the appearance of the root after 10 hours? Draw and label a diagram to visualize your answer. (2 marks)

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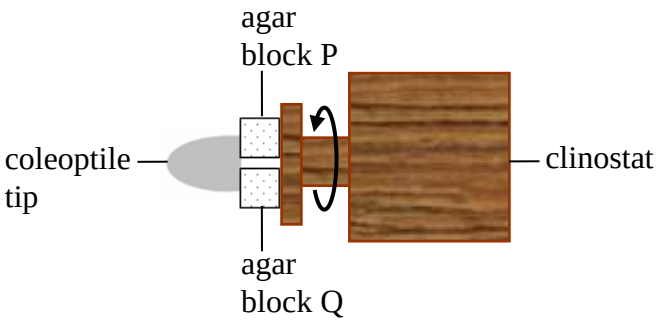
(c) Two other coleoptile tips were treated in the following ways before they were placed on the decapitated coleoptile as in the above experiment. Describe and explain what you would observe after 10 hours.

(i)



(3 marks)

(ii)



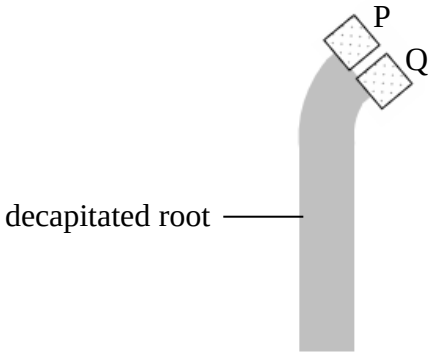
(3 marks)

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##

(a) In Figure I, more auxins moved downwards due to gravity. (1)
More auxins diffused into agar block Q than into P. (1)
Hence, the side of the coleoptile under Q grew faster than the side under P. (1)

(b) Correct direction of growth (1)
Correct labeling (1)



(c) (i) The decapitated coleoptile would grow vertically without bending. (1)
The mica plate prevented auxins from moving downwards due to gravity. (1)
Hence, auxins distribute evenly in P and Q. (1)

(ii) The decapitated coleoptile would grow vertically without bending. (1)
The effect of gravity on auxin distribution was cancelled out by the rotation of the clinostat. (1)
Hence, auxins distribute evenly in P and Q. (1)

(11 marks)

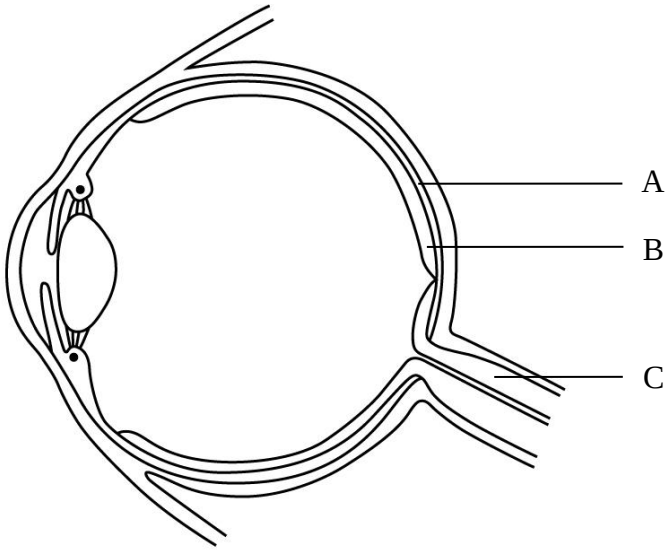
##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)	Applying concepts	1
	Presenting information	1
(c)(i)	Applying concepts	3
(c)(ii)	Applying concepts	3

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|!|BCELC0160000002|!|

The diagram below shows a section through a human eye.



(a) Name structures A, B and C. (3 marks)

(b) A student wrote ‘messages travel along structure C’.

(i) State the precise term for these ‘messages’. (1 mark)

(ii) Where are these messages carried to? (1 mark)

(c) Detachment of layer B from layer A can cause blindness. Explain this with reference to the function of layer A. (4 marks)

##

(a) A: choroid (1)
B: retina (1)
C: optic nerve (1)

(b) (i) nerve impulses (1)
(ii) visual centre / cerebrum (1)

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(c) Layer A contains blood vessels which supply nutrients / oxygen (1)
to the photoreceptors in layer B. (1)
If layer B detaches from layer A, the photoreceptors will die. (1)
Light falling on layer B cannot be detected / no nerve impulses can be produced. (1)
Thus, the vision is impaired.

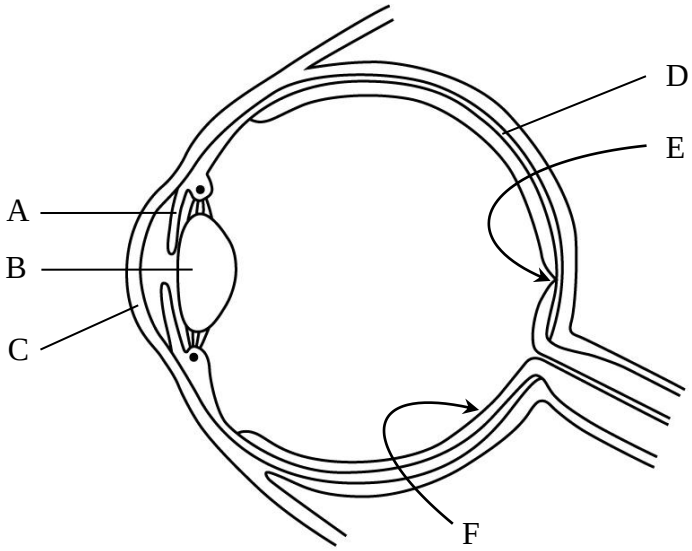
(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	1
(c)	Applying concepts	4

|!|BCELA0160000003|!| ★★

The diagram below shows a section through a human eye.



(a) Using the letters in the diagram, state all the parts that are transparent. (1 mark)

(b) State ***one*** function of structure D. How is structure D adapted to this function? (2 marks)

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(c) Describe the change in structure B when a person is watching a distant object moving towards him. Explain how this change is brought about. (3 marks)

(d) A person tries to look at a faint star in the sky at night. The image of the star may fall on E or F. Compare and explain what the person sees when the image falls on each of the two spots. (4 marks)

##

(a) B, C (1)

(b) Any ***one*** set of the following: (2)
To supply nutrients / oxygen to the eyeball (1)
as it is richly supplied with blood / capillaries. (1)
OR
To prevent reflection of light within the eye (1)
as it contains dark pigments. (1)

(c) Structure B would become thicker / more covex. (1)
This is caused by the contraction of the circular ciliary muscles (1)
which leads to a decrease in tension in the suspensory ligaments. (1)
The lens becomes thicker due to its elasticity.

(d) Any ***one*** set of the following: (4)
The star can be seen only when its image falls on F. (1)
This is because the light intensity from the star is so low (1)
that it is only sufficient to stimulate the rod cells (1)
which can be found at F but are absent at E. (1)
OR

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F contains rod cells while E contains only cone cells. (1)
Since the light intensity from the star is low (1)
the cone cells will be stimulated to a smaller extent than the rod cells. (1)
Therefore, when the image falls on F, the star is seen to be brighter than when it falls on E. (1)

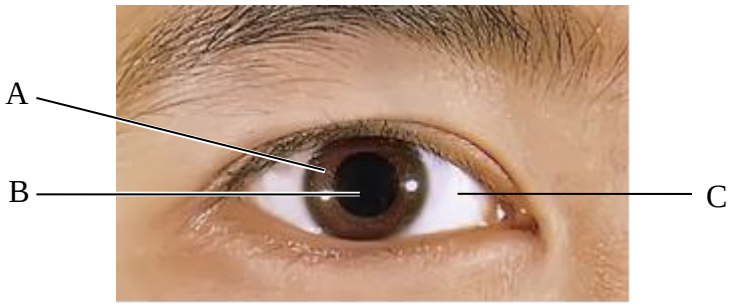
(10 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)	Applying concepts	3
(d)	Applying concepts	4

|!|BCELB0160000004|!| ★

The photograph below shows a person's eye.



(a) Name structures A, B and C. (3 marks)

(b) State the change in structure B when a person walks out from a dark room into bright daylight. Explain the significance of this change. (3 marks)

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(c) Describe how the eye and the nervous system work together to bring about the reflex in (b). (4 marks)

##

(a) A: iris (1)
B: pupil (1)
C: sclera (1)

(b) Structure B constricts / becomes smaller. (1)
This reduces the amount of light entering the eye (1)
so as to prevent over-stimulation / damage of the photoreceptors / retina. (1)

(c) The increase in light intensity stimulates the photoreceptors in the retina to generate nerve impulses. (1)
Nerve impulses are transmitted through sensory nerve fibres / the optic nerve to the brain. (1)
The brain sends nerve impulses to the iris through moter nerve fibres, (1)
causing the contraction of the circular muscles of the iris. (1)

(10 marks)

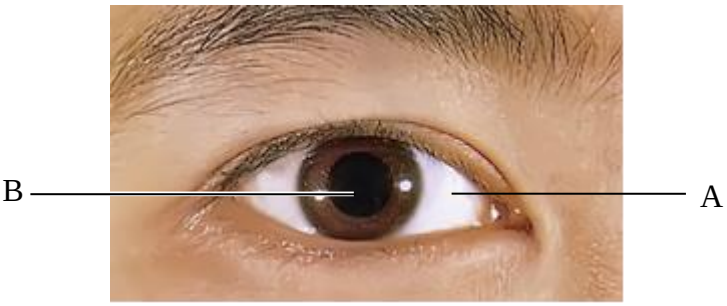
##

	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)	Understanding basic concepts	3
(c)	Applying concepts	4

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|!|BCELB0160000005|!| ★ **Cross topic**

The photograph below shows a person's eye.



(a) Describe how structure A is adapted to its function. (2 marks)

(b) Explain why structure B is black in colour. (2 marks)

(c) State the change in structure B when a person enters a dark room from a bright corridor. Describe how the change is brought about. (3 marks)

(d) The change describe in (c) is a reflex action. State ***two*** differences between reflex actions and voluntary actions. (2 marks)

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##

(a) It is tough (1)
and so it can protect the inner structures / maintain the shape of the eyeball. (1)

(b) As all the light entering the eye is absorbed by the retina and the choroid, (1)
no light is reflected out of the eye through the pupil. (1)

(c) Structure B dilates / becomes larger. (1)
This is brought about by the contraction of the radial muscles of the iris (1)
and the relaxation of the circular muscles. (1)

(d) Any ***two*** of the following: (2)

- Reflex actions do not involve the cerebrum while voluntary actions involve the cerebrum.
- Reflex actions are inborn while voluntary actions are not inborn (require learning).
- Reflex actions are usually faster in response than voluntary actions.
- Reflex actions are stereotypes while voluntary actions are not.

(or other reasonable answers)

(9 marks)

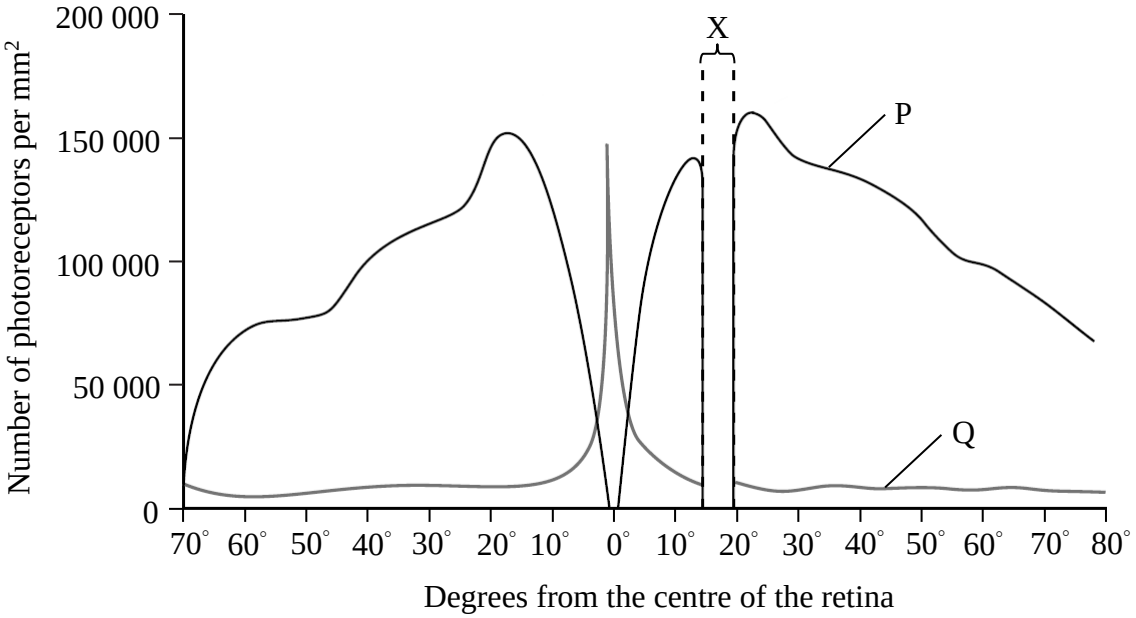
##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Applying concepts	2
(c)	Understanding basic concepts	3
(d)	Critical thinking	2

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|!|BCELB0160000006|!| ★

The graph below shows the distribution of two types of photoreceptors, P and Q, across the retina of a human eye.



(a) Name the photoreceptors represented by P and Q. (2 marks)

_____

_____

(b) A person cannot see an object if its image falls on X. Explain why. (2 marks)

_____

_____

(c) In a dim room, a man saw a cockroach at the edge of his vision. When he moved his eyes to look straight at it, he could no longer see it. Use the information from the graph to explain

(i) why he could see the cockroach at the edge of his vision. (3 marks)

_____

_____

_____

_____

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(ii) why he could not see the cockroach when he looked straight at it. (3 marks)

##

(a) P: rod cells (1)
Q: cone cells (1)

(b) There are no photoreceptors to detect the image. (1)
No nerve impulses would be sent to the brain. (1)

(c) (i) The image of the cockroach falls on the periphery of the retina (1)
which has a larger number of rod cells than cone cells. (1)
Rod cells can be stimulated by low light intensities / in dim light. (1)
Thus, the man could see the cockroach at the edge of his vision.

(ii) When looking straight at the cockroach, the image of the cockroach falls on the centre of the retina (yellow spot) (1)
which contains cone cells only. (1)
Cone cells are not stimulated by low light intensities / need bright light to function well. (1)
Therefore, the man could not see the cockroach when he looked straight at it.

(10 marks)

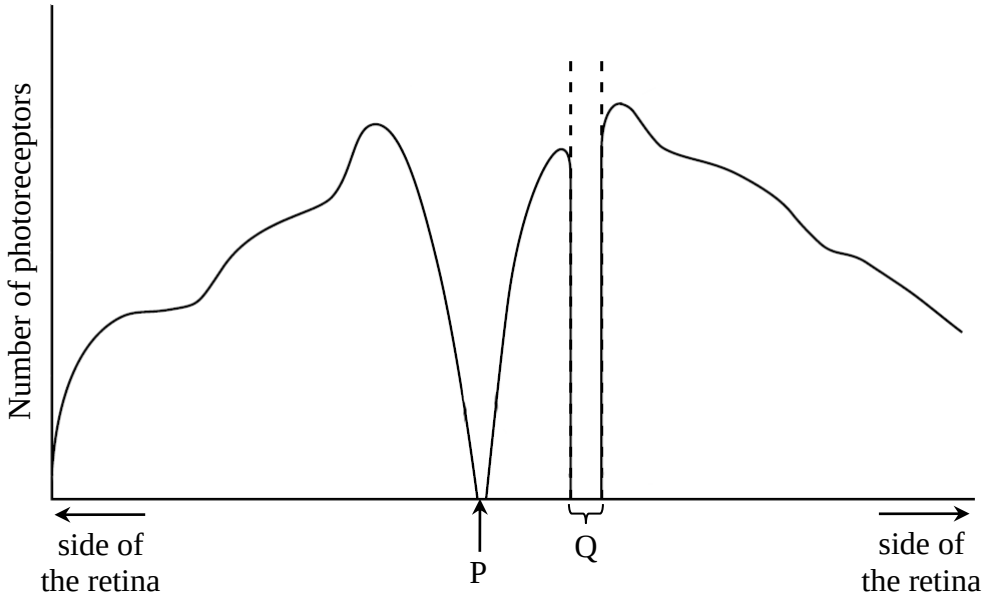
##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Applying concepts	2
(c)(i)	Applying concepts	3
(c)(ii)	Applying concepts	3

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|!|BCELB0160000007|!| ★

The graph below shows the distribution of rod cells across the retina of a human eye.



(a) Name the regions of the retina labelled P and Q. (2 marks)

(b) Draw a line on the graph to show the distribution of cone cells in the retina. (2 marks)

(c) State ***two*** ways, other than shape and distribution, in which rod cells differ from cone cells. (2 marks)

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(d) How can an image be perceived after rod cells and cones cells are stimulated?

(3 marks)

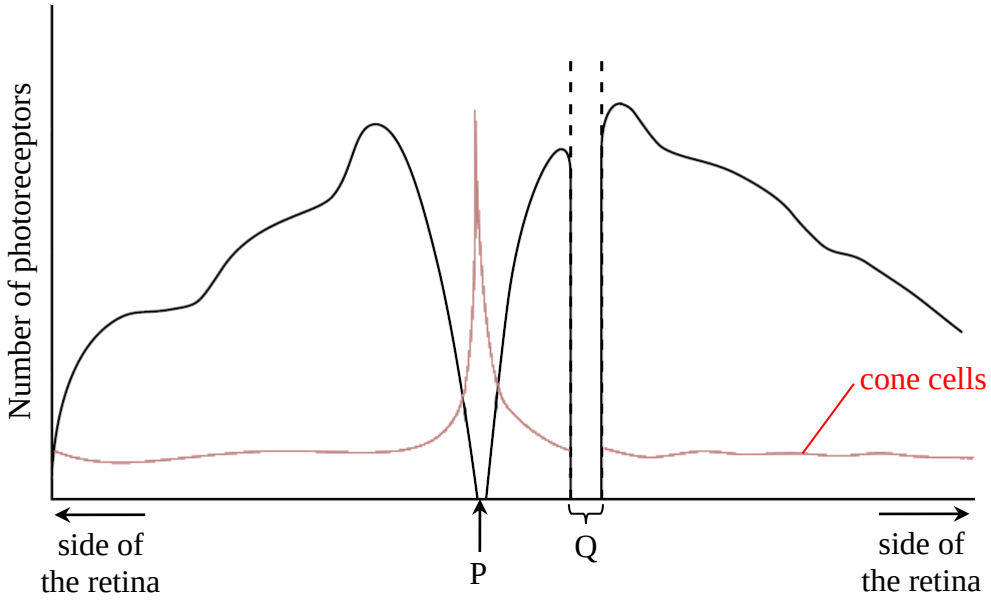
##

(a) P: yellow spot (1)

Q: blind spot (1)

(b) Greatest number at P. (1)

None at Q. (1)



(c) Any ***two*** of the following: (2)

- Rod cells are more sensitive to light / respond to lower light intensities.
- Rod cells cannot detect colours while cone cells can detect colours.
- There is only one type of rod cells while there are three types of cone cells.

(d) When rod cells and cone cells generate nerve impulses after they are stimulated by light. (1)

Nerve impulses are transmitted along the optic nerve (1)

to the visual centre in the brain / cerebrum for interpretation. (1)

(9 marks)

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##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Presenting information	2
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	3

|!|BCELB0160000008|!| ★

(a) After reading a book for some time, a man felt his eyes were tired. Explain why. (1 mark)

(b) The man looked up from the book to a clock on the wall.

(i) Describe and explain the changes that occurred in his eyes. (3 marks)

(ii) It was 3:00 p.m. Draw the image of the two hands of the clock formed on the retina of his eyes. (1 mark)

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(iii) Describe how the image on the retina generates vision in the brain. (4 marks)

##

(a) The ciliary muscles had been contracting for too long. (1)

(b) (i) The ciliary muscles relaxed. (1)

The suspensory ligaments became tightened / increased in tension. (1)

The lens was pulled to become thinner / less convex. (1)

(ii)



(1)

(iii) Photoreceptors in the retina were stimulated. (1)

They generated nerve impulses (1)

which are transmitted along the optic nerve (1)

to the visual centre in the brain / the cerebrum fro interpretation into vision. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)(i)	Applying concepts	3
(b)(ii)	Presenting information	1
(b)(iii)	Understanding basic concepts	4

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|!|BCELC0160000009|!|

Candy was sitting for an examination in the school hall. Though she could clearly see the clock at the far end of the school hall, she could not see her watch clearly without putting on her glasses.

(a) What kind of eye defect did Candy probably have? (1 mark)

(b) State ***two*** features of the eyeball that might have caused this eye defect. (2 marks)

(c) Draw a ray diagram to show how this eye defect is corrected by wearing glasses. (3 marks)

(d) Candy found it difficult to distinguish colours in dim light. Is this situation normal? Explain why. (3 marks)

##

(a) Long sight (1)

(b) The eyeball being too short (1)

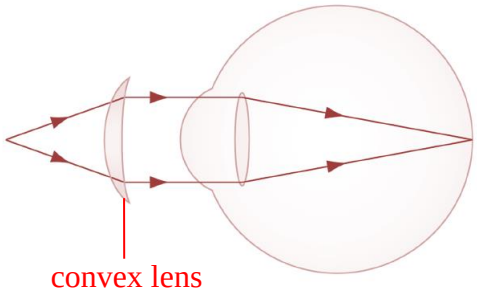
The lens being too thin (1)

(c) Convex lens correctly labelled (1)

Light rays from a nearby object converged by the convex lens (1)

Light rays intersect on the retina (1)

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(d) Yes, this is normal. (1)
Cone cells, which are responsible for detecting colours (1)
do not function well in dim light. (1)

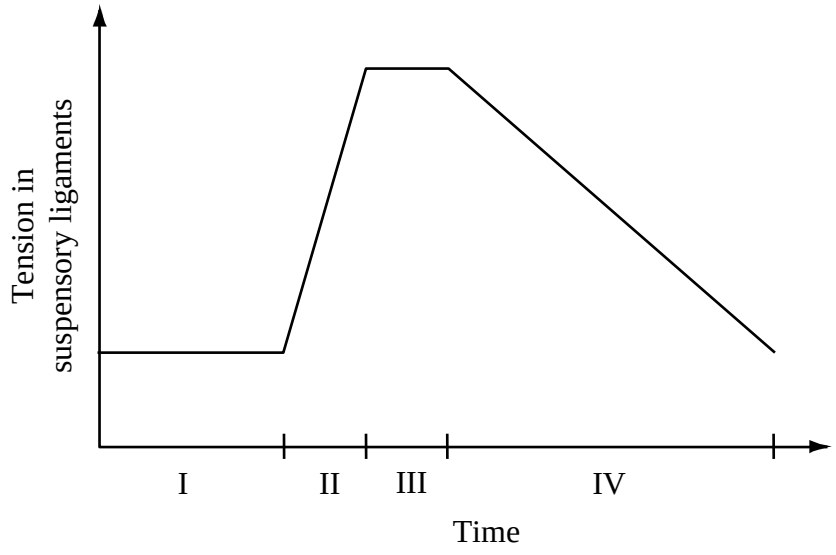
(9 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Understanding basic concepts	2
(c)	Presenting information	3
(d)	Applying concepts	3

|!|BCELB0160000010|!| ★

The graph below shows the changes in the tension in the suspensory ligaments of a person's eye in a period of time. During the period, the eye was focusing on an object.



(a) Complete the table below, which describes the positions of the object in different time period. (4 marks)

Time period	Position of the object
	The object was furthest from the man.
	The object was nearest to the man.
	The object was moving away from the man.
	The object was moving towards the man.

(b) State the change in the curvature of the lens during period IV. Describe how this change was brought about. (3 marks)

(c) The man was short-sighted.

(i) Draw a ray diagram in the space below to show the pathway of light entering the eye of the man when he was looking at a distant object. (3 marks)

(ii) How could this eye defect be corrected? (1 mark)

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##

(a)

Time period	Position of the object
III	The object was furthest from the man.
I	The object was nearest to the man.
II	The object was moving away from the man.
IV	The object was moving towards the man.

(4)

(b) The curvature of the lens was increasing. (1)

The circular ciliary muscles contracted. (1)

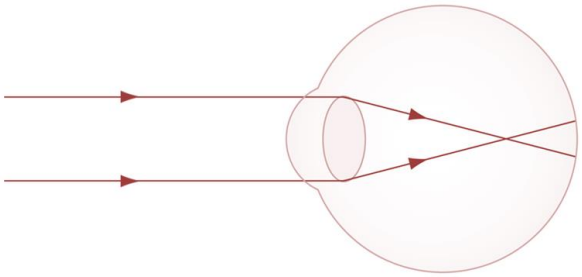
This reduced the tension in the suspensory ligaments. (1)

Thus, the lens became more convex due to its elasticity.

(c) (i) Correct outline of the eyeball and lens (1)

Parallel light rays enter the eye (1)

Light rays intersect in front of the retina and extend to the retina (1)



(ii) Wearing concave lenses / Undergoing surgery (1)

(11 marks)

##

	Skills assessed	Marks
(a)	Analysing data	4
(b)	Understanding basic concepts	3
(c)(i)	Presenting information	3
(c)(ii)	Understanding basic concepts	1

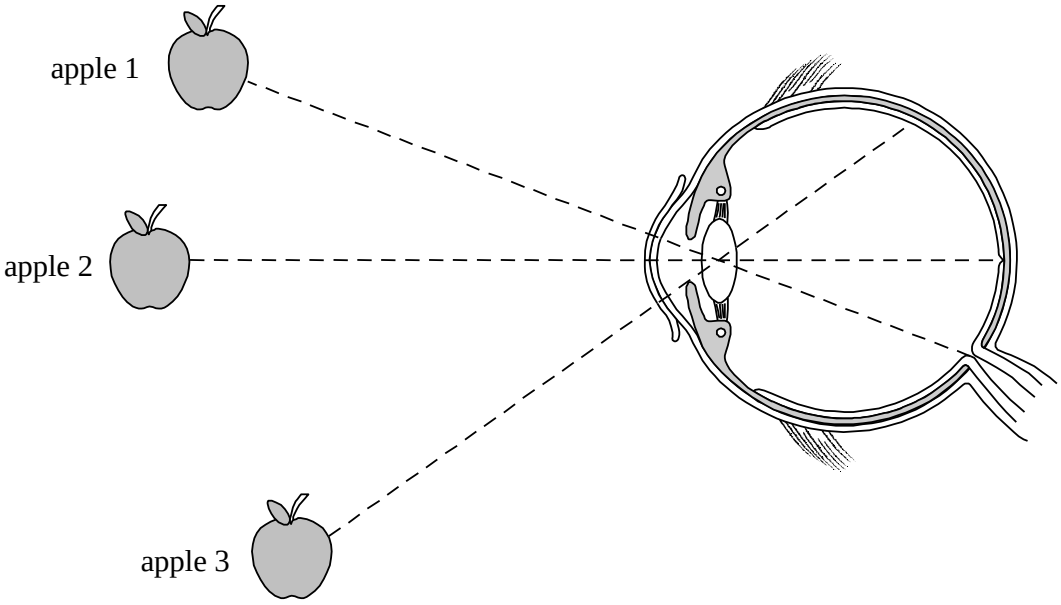
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|!|BCELB0160000011|!| ★

Adam was having dinner with his family in the bright dining room. His mother asked him to go to the kitchen to take an apple. The kitchen was dimly lit.

(a) Describe how his eyes adjusted to the dim kitchen light? (3 marks)

(b) Adam switched on the light in the kitchen to look for an apple. Three apples were placed on a table in the kitchen. The diagram below shows a section of Adam's eye and the light rays from the three apples.



(i) Which of the three apples would be perceived most clearly? Explain your answer. (3 marks)

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(ii) When Adam's eye kept focusing in the direction of apple 2, he could see apple 1 but could not detect its colour. Give an explanation for this. (3 marks)

##

(a) The radial muscles of the iris contracted (1)
and the circular muscles are relaxed. (1)
The pupil dilated / became larger. (1)

(b) (i) Apple 2 (1)
The image of apple 2 fell on the yellow spot, (1)
which contains the greatest number of cone cells to detect the image. (1)

(ii) Because the image of apple 1 fell on the peripheral part of the retina, (1)
which has rod cells to detect light from the apple (1)
but there are few cone cells and so the colour of the apple cannot be detected effectively. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)(i)	Analysing data	3
(b)(ii)	Analysing data	3

|!|BCELA0160000012|!| ★★ **Cross topic**

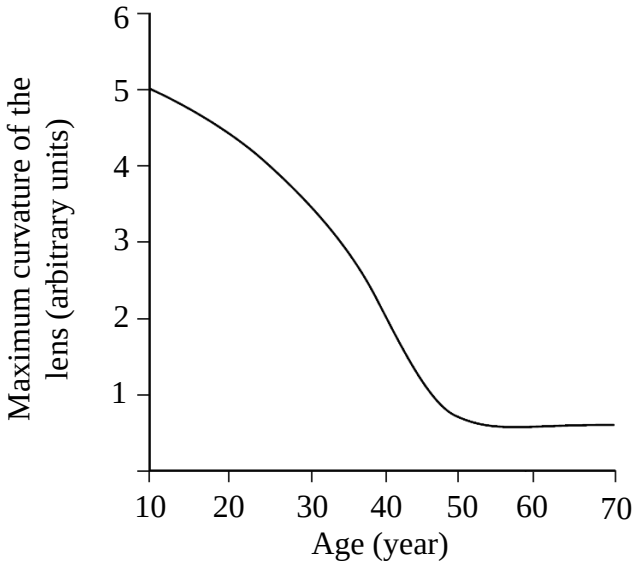
(a) The iris of the eye contains antagonistic muscles which control the size of the pupil.

(i) Use your knowledge of the muscles of the iris to explain the action of antagonistic muscles. (3 marks)

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(ii) The pupil response to light intensity is a reflex action coordinated by the brain. Draw a flowchart to show the nervous pathway involved. (3 marks)

(b) When focusing on objects at different distances, the curvature of the lens in the eye changes. The graph below shows the maximum curvature of the lens in people of different ages.



Using the information from the graph to explain how the ability of the lens to change its curvature affects the ability to focus clearly as people get older.

(3 marks)

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##

(a) (i) Antagonistic muscles work in pair to bring about opposing actions. (1)
The circular muscles of the iris contract and the radial muscles relax to reduce the size of the pupil. (1)
The radial muscles of the iris contract and the circular muscles relax to increase the size of the pupil. (1)

(ii) photoreceptors → sensory neurone / optic nerve → interneurone in the brain → motor neurone / nerve → iris muscle (3)

(b) As people get older, the lens becomes less elastic. (1)
When looking at a near object, the lens is not concave enough (1)
and fails to converge light to form a clear image on the retina. (1)

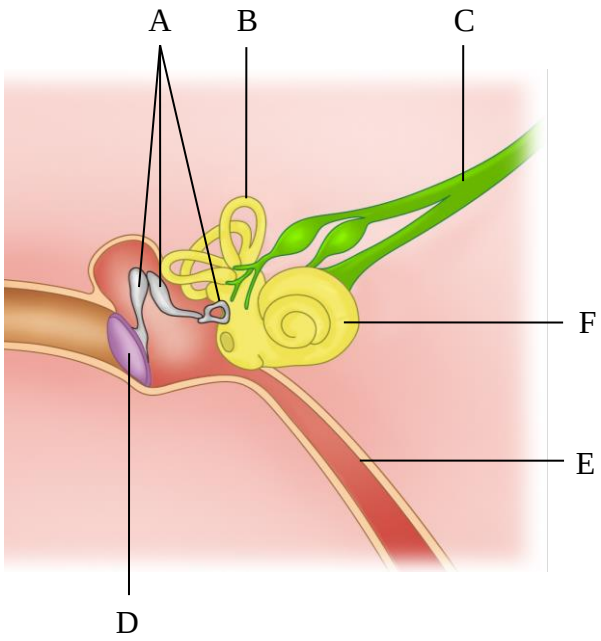
(9 marks)

##

	Skills assessed	Marks
(a)(i)	Applying concepts	3
(a)(ii)	Presenting information	3
(b)	Analysing data	3

|!|BCELA0160000013|!| ★★

The diagram below shows part of the human ear.



(a) Using the letters in the diagram, state all the parts, in the correct sequence, involved in hearing. (2 marks)

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(b) Using the letters in the diagram, state all the parts that can vibrate. (1 mark)

(c) Using the letters in the diagram, state all the parts that are filled with liquid. (1 mark)

(d) A major cause of tinnitus (ringing in the ears) is prolonged exposure to loud noise. Name the cells in the inner ear that are likely to be damaged by loud noise and suggest why this may lead to tinnitus. (2 marks)

(e) Suggest how the middle ear may be infected when a person suffers from a sore throat. (2 marks)

##

(a) D → A → F → C (2)

(b) D, A (1)

(c) B, F (1)

(d) Sensory hair cells (1)
The damage causes the sensory hair cells to generate nerve impulses continuously. (1)

(e) Pathogens from the throat may enter the middle ear (1)
through the Eustachian tube. (1)

(8 marks)

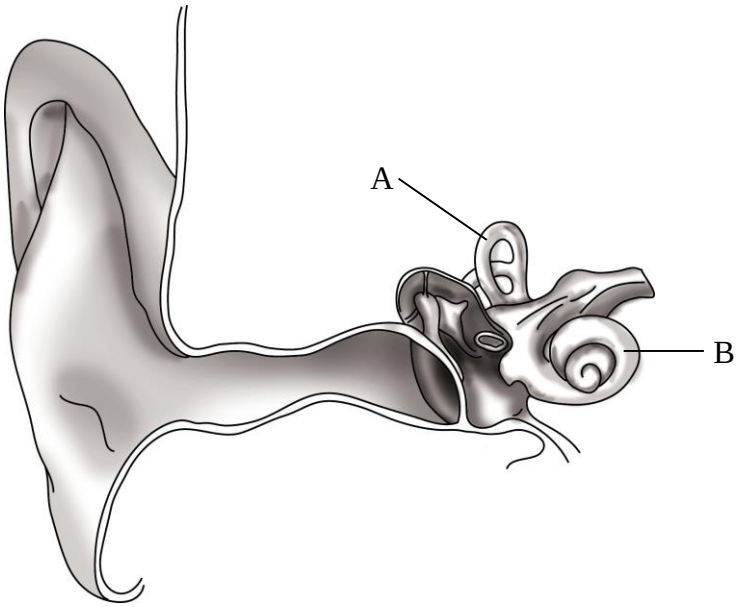
##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Applying concepts	2
(e)	Applying concepts	2

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|!|BCELB0160000014|!| ★

The diagram below shows the human ear and its associated structures.



(a) Name structure A and state its function. (2 marks)

(b) Name structure B and state its function. (2 marks)

(c) (i) As an airplane is descending to land, passengers may feel pain in the ears and cannot hear properly. Explain why. (3 marks)

(ii) Describe how swallowing and yawning may help relieve ear pain. (3 marks)

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##

(a) Semicircular canals (1)

They detect the direction of head movements. (1)

(b) Cochlea (1)

It detects sound in the form of liquid movements. (1)

(c) (i) When the airplane is descending, the atmospheric pressure increases rapidly and become higher than the pressure inside the middle ear. (1)

The eardrum is pushed inwards, causing ear pain. (1)

The bulged eardrum cannot vibrate freely and so hearing is affected. (1)

(ii) Swallowing and yawning opens the Eustachian tube (1)

to allow air to enter the middle ear. (1)

This equalizes the pressure on both sides of the eardrum (1)

so that the eardrum can return to its normal shape.

(10 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	2
(c)(i)	Applying concepts	3
(c)(ii)	Applying concepts	3

|!|BCELB0160000015|!| ★ **Cross topic**

(a) Reflex actions and tropic responses are important responses in humans and plants respectively. Give ***three*** differences the two types of responses. (3 marks)

(b) Name the group of substances responsible for causing tropic responses in plants and describe their effect. (2 marks)

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(c) Describe how shoots and roots respond to unilateral light. Explain the significance of each response. (4 marks)

##

(a) Any ***three*** pairs of the following: (3)

Reflex actions	Tropic responses
• elicited by external or internal stimuli	• elicited by external stimuli only
• stimuli involved may or may not be unilateral	• stimuli involved are unilateral
• messages involved are electrochemical in nature	• messages involved are chemical in nature
• most are non-directional responses	• directional responses
• quick responses	• slower responses
• non-growth responses	• growth responses

(b) Auxins (1)

They affect plant growth by affect the rate of cell elongation. (1)

(c) Shoots grow towards the unilateral light / are positively phototropic. (1)

This brings the leaves to face the light source so that they can obtain the maximum amount of light for photosynthesis. (1)

Roots grow away from unilateral light / are negatively phototropic. (1)

This ensures that the roots grow downwards into the soil for anchorage. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Critical thinking	3
(b)	Understanding basic concepts	2
(c)	Applying concepts	4

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|!|BCELC0160000016|!| **Cross topic** **SI**

In an investigation, some mung beans were allowed to germinate and grow in a light-proof box with a slit on one side. The photograph below shows the appearance of the seedlings after three days.



(a) Name the growth response of the seedlings demonstrated in this investigation. (1 mark)

(b) Explain the growth response of the seedlings with respect to the effects of auxins. (3 marks)

(c) In order to show that the growth response of the seedlings is due to the unilateral light, a control should be included.

(i) How you would set up a control for this investigation? (1 mark)

(ii) What would be the expected results of the control? (1 mark)

(d) Suggest why the seedlings are yellow instead of green. (2 marks)

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##

(a) Phototropism (1)

(b) Unilateral light causes auxins to be transported from the illuminated side to the shaded side of the shoots. (1)

The shaded side contains a higher concentration of auxins. (1)

Thus, the shaded side grows faster than the illuminated side and the shoots bend towards the light. (1)

(c) (i) Placing the seedlings on a rotating clinostat. (1)

(ii) The shoots would grow straight upwards. (1)

(d) The seedlings cannot produce enough chlorophyll (1)

due to a lack of magnesium / nitrogen / sunlight. (1)

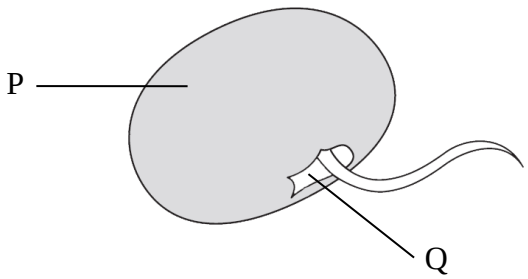
(8 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Applying concepts	3
(c)(i)	Practical skills	1
(c)(ii)	Practical skills	1
(d)	Applying concepts	2

|!|BCELC0160000017|!| **Cross topic** **SI**

(a) The diagram below shows a germinating mung bean.

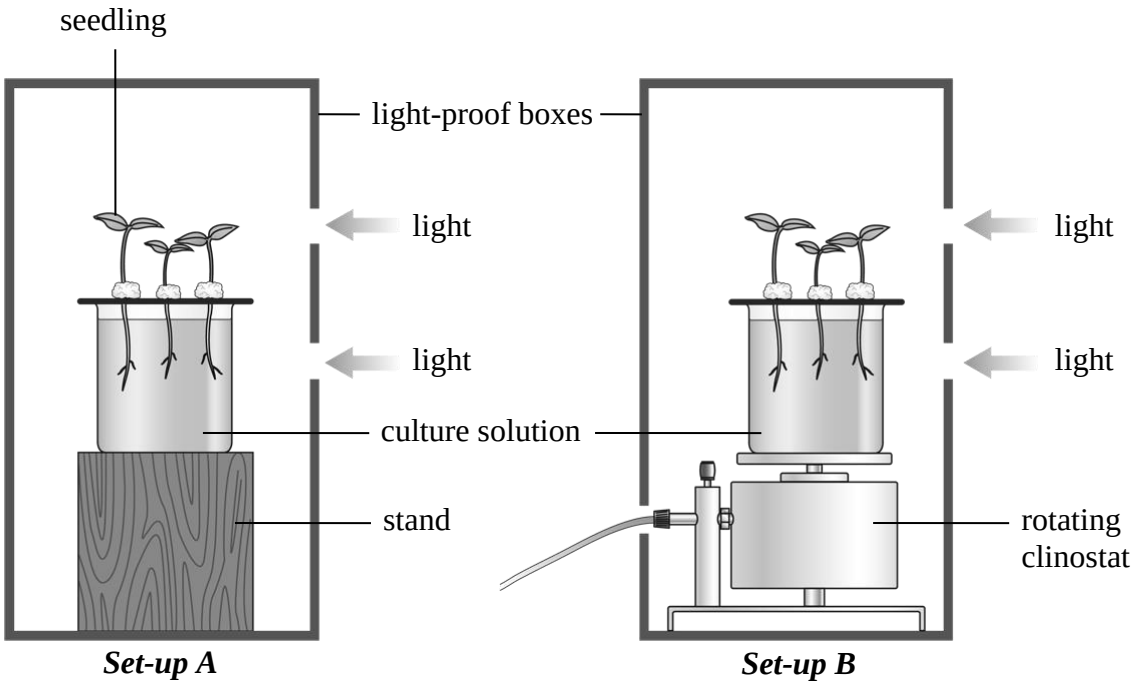


(i) Name structure P and state its functions. (2 marks)

(ii) Name structure Q and state its functions. (2 marks)

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(b) The diagram below shows two set-ups used to investigate phototropism of mung bean seedlings.



(i) Describe the appearance of the shoots and the roots of the seedlings in set-up A after two days. (2 marks)

(ii) Explain the results stated in (i) on a hormonal basis. (4 marks)

(iii) Explain the purpose of set-up B. (2 marks)

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##

(a) (i) Seed coat (1)

It protects the embryo from damage / attack by microorganisms. (1)

(ii) Cotyledon (1)

It contains stored food for the growth of the plumule and the radicle during seed germination. (1)

(b) (i) The shoots would grow and bend towards the unilateral light, (1)

while the roots would grow and bend away from the unilateral light. (1)

(ii) Unilateral light causes auxins to be transported from the illuminated side to the shaded side. (1)

There are more auxins on the shaded side. (1)

The high concentration of auxins promotes shoot growth on the shaded side and so the shoots bend towards the unilateral light. (1)

The high concentration of auxins inhibits root growth on the shaded side and so the roots bend away the unilateral light. (1)

(iii) The rotating clinostat cancels out the effect of unilateral light. (1)

Set-up B is used as a control to show that any growth responses observed in set-up A are due to unilateral light. (1)

(12 marks)

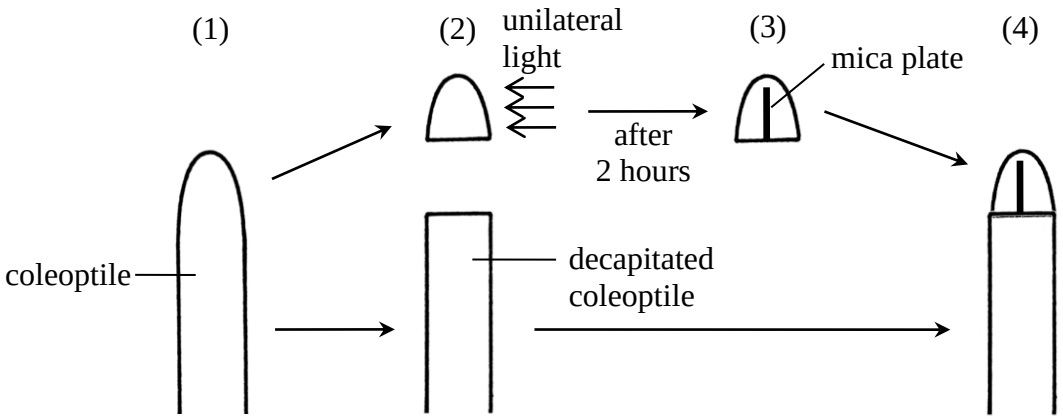
##

	Skills assessed	Marks
(a)(i)	Understanding basic concepts	2
(a)(ii)	Understanding basic concepts	2
(b)(i)	Analysing data	2
(b)(ii)	Applying concepts	4
(b)(iii)	Practical skills	2

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|!|BCELB0160000018|!| ★ **SI**

The diagram below shows an experiment in which a student cut the tip off a coleoptile and exposed it to unilateral light. After two hours, he inserted a mica plate into the cut tip and placed the cut tip back on top of the decapitated coleoptile. The set-up was left undisturbed in the dark for two days.



(a) What change in the decapitated coleoptile would be observed after two days? (1 mark)

(b) Account for the change in the coleoptile. (4 marks)

(c) What would happen if the mica plate were inserted into the cut tip before exposing the cut tip to the unilateral light? Explain your answer. (3 marks)

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(d) What would happen if the coleoptile were replaced by a radicle in step (4)? Explain your answer. (2 marks)

_____

_____

##

(a) The decapitated coleoptile would bend towards the right. (1)

(b) In step (2), the unilateral light caused auxins to be transported from the illuminated (right hand) side of the cut tip to the shaded (left hand) side. (1)
When the cut tip was placed back onto the decapitated coleoptile in step (4), the higher concentration of auxins on the left hand side (1)
promoted the growth of the coleoptile. (1)
Thus, the left hand side of the coleoptile would grow faster than the right hand side. (1)

(c) The coleoptile would grow straight upwards / would not bend. (1)
This is because the mica plate would block the transport of auxins from the illuminated side to the shaded side (1)
and auxins would be evenly distributed in the cut tip. (1)

(d) The radicle would bend towards the left hand side (1)
because the high concentration of auxins on that side would inhibit the growth of the radicle. (1)

(10 marks)

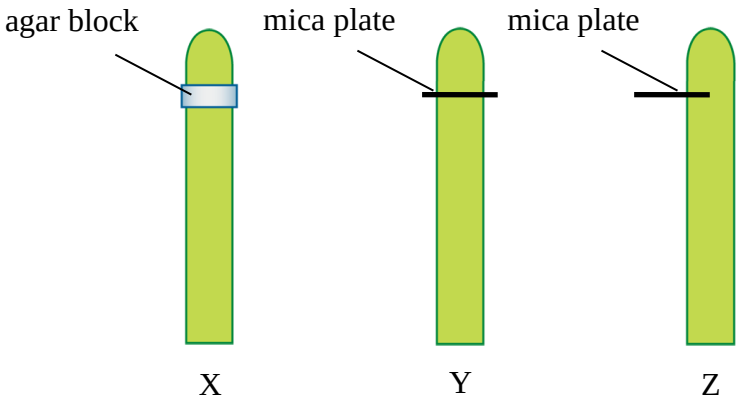
##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Applying concepts	4
(c)	Practical skills	3
(d)	Applying concepts	2

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|!|BCELB0160000019|!| ★ SI

The diagram below shows three oat coleoptiles treated in different ways. The coleoptiles are kept in the dark.



(a) Why are the coleoptiles kept in the dark? (1 mark)

(b) Describe how the coleoptiles will grow (if any) after two days. Explain your answer in each case. (6 marks)

(c) Suggest a control set-up for this experiment the show that the growth response of the coleoptiles is caused by some chemical substances produced at the tip. (1 mark)

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##

(a) To eliminated the effect of light on the growth of the coleoptiles. (1)

(b) X: The coleoptile will grow straight upwards (1)
because auxins from the tip can diffuse through the agar block and are transported to the region of elongation to stimulate growth of the coleoptile. (1)
Y: The coleoptile will not grow (1)
because auxins from the tip cannot diffuse through the mica plate to the region of elongation. (1)
Z: The coleoptile will grow and bend towards the left (1)
because the mica plate blocks the transport of auxins down the left side / auxins from the tip are transported down the right side of the coleoptile. The right side with a higher auxin concentration grows faster than the left side. (1)

(c) Use a decapitated coleoptile and put an agar block on its top. (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Practical skills	1
(b)	Analysing data	6
(c)	Practical skills	1

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|!|BCELB0160000020|!| ★ **SI**

In an investigation, a number of shoot tips were cut off from seedlings. Each tip was placed on two agar blocks, X and Y, and subjected to unilateral light as shown in Figure 1. After several hours, the agar blocks were place on the cut ends of decapitated shoots and roots as shown in Figure 2. The results are observed after 24 hours.

Figure 1

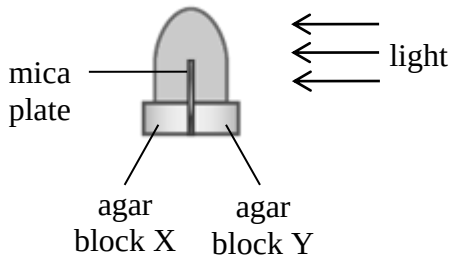
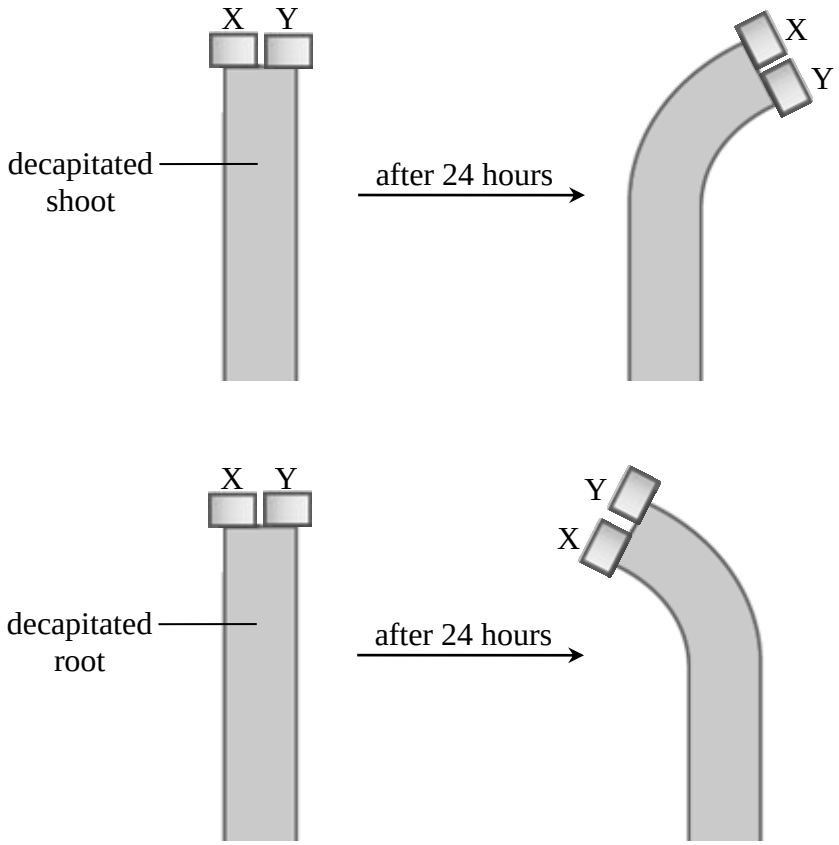


Figure 2



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(a) Explain the result of the decapitated shoot on a hormonal basis. (4 marks)

(b) Explain why a different response was shown by the decapitated root. (2 marks)

(c) State and explain ***one*** precaution that should be taken in this investigation.(2 marks)

##

(a) The unilateral light caused auxins in the tip to be transported from the illuminated side to the shaded side. (1)
More auxins diffused into agar block X than into agar block Y. (1)
When the agar blocks were placed on the cut ends of the decapitated shoot, more auxins are transported to the region of elongation down the left side of the shoot.(1)
The left side with a higher auxin concentration grew faster than the right side and so the shoot bent towards the right. (1)

(b) The higher concentration of auxins on the left side inhibited root growth. (1)
Thus, the left side of the root grew more slowly than the right side. (1)

(c) The decapitated shoots and roots should be kept in the dark (1)
to eliminate the effect of light on the distribution of auxins. (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Analysing data	4
(b)	Analysing data	2
(c)	Practical skills	2

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